Ohm’s Law: Four Brief PhET Simulations

Voltage, current and resistance, related

> SYNOPSIS
>
> You have explored the three main electrical quantities required to characterize electrical circuits: voltage, current, and resistance. By now, you should have a clear picture / understanding / description of voltage, current, and resistance. These three quantities are intimately related, as given by Ohm’s Law. Ohm’s law is simple yet incredibly powerful, underpinning the understanding of all electrical circuits you will ever encounter!

What Ohm’s law says

Ohm’s Law is

$$V = IR$$

where V is the voltage in volts (V), I is the current in amps (A), and R is the resistance in ohms (Ω, capital Greek letter omega).

It simply says that when an electrical potential difference is high (“high pressure”), there will be more current pushed through (“bigger volume of water passing through a point”) for a given resistance (“diameter of pipe *restriction*” or “bottleneck”). So, for a fixed resistance (“bottleneck diameter”), there is a direct relationship (one is directly proportional to another) between voltage and current: when the voltage doubles (triples and so on), the current doubles (triples and so on) or vice versa.

If you plot voltage (y-axis) vs. current (x-axis), it is a straight line going through the origin with a slope given by the resistance. Or, if you plot the current (y-axis) vs. voltage (x-axis), it is a straight line going through the origin with a slope given by the inverse of resistance ($1/R$).

What would happen to the current if you increased the resistance when the voltage was fixed (this is quite common as you can switch out different resistors while using the same battery)? What happens if you decrease the bottleneck pipe diameter in a water pipe for the same water pump pressure? The circuit’s current will decrease, just like the water flow would decrease with a more severe bottleneck. Therefore, plotting current (y-axis) vs. resistance (x-axis) is an inverse curve showing decreasing current with increasing resistance.

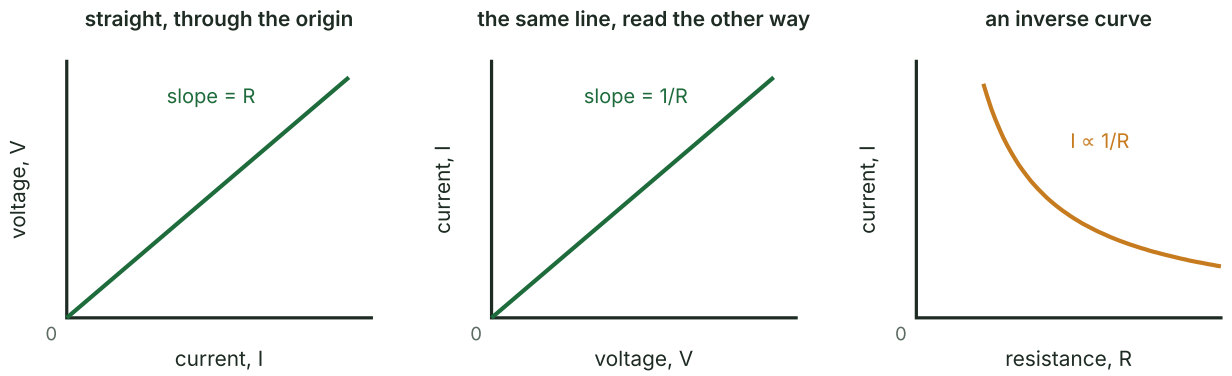


Figure 1. The three plots named in the paragraphs above. The first two are the same straight line read two ways, which is why one slope is the inverse of the other. The third is not a line at all.

You will explore Ohm's Law in detail in this lab using the following four brief PhET simulations:

1. Battery voltage. (search "battery voltage phet sim.")
2. Ohm's law. (search "ohms law phet sim.")
3. Battery-resistor circuit. (search "battery resistor circuit phet sim.")
4. Resistance in a wire. (search "resistance in a wire phet sim.")

PERFORMANCE EXPECTATIONS

1. Describe mathematical relationships between voltage, current, and resistance.
2. Use Ohm's law to describe the relationship between voltage, current, and resistance.
3. Describe the effect of voltage and resistance on the flow of electrons, and how friction between electrons and obstacles to electron flow generates heat in resistors.
4. Mathematically describe how resistance is related to the resistivity (material property), the length, and the crossectional area of the material.

Instructions and Questions

1) Run the *Battery Voltage* PhET simulation

Check the box "Show Battery."

i. Do the small blue spheres represent positive or negative charges?

ii. Which side of the battery is labeled positive, and which is negative?

iii. How can you determine which side of the battery is positive or negative just by the *location* of the blue charges?

iv. Complete the statement: “A battery is a device that has a built-in mechanism that causes electrons to pile up on one end, creating a ___ terminal where there is excess electrons and a ___ terminal where there is a lack of electrons. … This tendency or drive to move towards the region with a lack of electrons is thus the ‘pressure’ on electrons to flow towards the region with a lack of electrons and is, therefore, the potential ___ or v___.”

2) Run the *Ohm’s Law* PhET simulation

Remember, Ohm’s Law states $V = IR$.

i. Slide the lever for voltage up and down. What does the size of the voltage correlate with within the circuit (see the circuit diagram below the equation)?

ii. What does the change in the size of the voltage change in Ohm’s law equation?

iii. What does this change in the equation mean when voltage size changes? In other words, what physical change happens when voltage changes?

iv. When you change the voltage, does the resistance change? Explain your answer.

v. Slide the lever for resistance up and down. What does the size of the resistance correlate with within the circuit? (Hint: it has something to do with the resistor, which is the red cylindrical object.)

vi. What does the change in the size of the resistance change in Ohm’s law equation?

vii. What does this change in the equation mean when the resistance size changes? In other words, what physical change happens when the resistance changes?

viii. When you change the resistance, does the voltage change? Explain your answer.

3) Run the *Battery-Resistor Circuit* PhET simulation

Check the “Show Voltage Calculation” and “Show Inside Battery” boxes.

i. What are the small blue dots?

ii. How can you tell which side is positive or negative terminal?

iii. A battery can be thought of as the “pump” that forces electrons to pile up at one end. Explain how this creates the “pressure” on the electrons at one end and, thus, the potential difference / voltage. Use the graphics / animation.

iv. Check and uncheck “Show Cores.” What does this switch do? What are the big green dots?

v. Leaving the “Show Cores” box checked, slide the Resistance control left and right. The red cylindrical object is the resistor. What do you think are the big green dots, and how do they affect the electron flow (current)?

vi. Slide the Voltage control left and right.

a. What happens to the electrons? Use the animation to describe the effect.

b. What happens to the temperature of the resistor? Explain the change in temperature using the animation.

c. The current is measured in Amps. Using the animation, describe what happens to the current.

vii. When the current is high through the resistor, the temperature rises higher than when the current is low. Using your knowledge of the role of friction, explain why.

viii. When resistance is high, you would expect higher friction. Then, why can you create the highest temperature in the resistor by minimizing the resistance while maximizing the voltage? (Hint: can you have friction without the flow of electrons?)

4) Run the *Resistance in a Wire* PhET simulation

This simulation focuses on the origin of resistance in terms of the properties of the resistor. Specifically, we need to consider the crossectional area of the wire (therefore the diameter of the wire, “the pipe size”), the length of the wire (the distance the electrons have to traverse), and the material property known as resistivity (how easy or hard it is for electrons to move through — for example, in metals, which are conductors, electrons are a lot freer to move than in plastics, which are insulators). The point of this simulation is that resistance should be considered from the view of what helps the flow of electrons and what impedes the flow of electrons.

i. Slide the Length lever up and down. Why does the resistance increase with length? Explain using the graphics and what the changes in length mean for the electron flow through the material.

ii. Slide the Area lever up and down. Why does the resistance decrease with area? Explain using the graphics and what the changes in area mean for the electron flow through the material.

iii. Slide the resistivity (Greek letter called rho, which looks like the letter “p”) up and down. Why does the resistance increase with the resistivity increase? Explain using the graphics and what the resistivity changes mean for the material’s electron flow. (Hint: the black dots are the obstacles to electron flow in the materials, such as nuclei or other particles. Changing resistivity means changing the material of the resistor, say from a metal (sparse dots) to an insulator (dense dots).)

Looking ahead

Note 1 named voltage, current and resistance and gave you a picture for them. This note has tied the three together with one equation and shown you, in the simulations, what each one *is* inside the resistor: more batteries, more pressure; more black dots, more collisions. **What comes next is circuits with more than one resistor in them**, where the same law has to be applied piece by piece rather than all at once.